Optimization of DTC-Based and Harmonic-Mixer-Based Fractional-N PLLs: Comparative Analysis of Jitter and Power Trade-Offs

Yuyang Zhu, *Graduate Student Member, IEEE*, Masaru Osada, *Member, IEEE*, Haoming Zhang, *Graduate Student Member, IEEE*, and Tetsuya Iizuka, *Senior Member, IEEE*

***Abstract*— As phase-locked loop (PLL) architectures become increasingly complex, optimizing the jitter and power performance through calculation alone is becoming more challenging for fractional-N PLLs. To find the most suitable PLL architecture that meets the jitter-power requirements of various applications, a simple and widely-applicable method is in demand to find the optimal jitter-power relation of different PLL architectures. In this paper, we propose the use of a multi-objective evolutionary algorithm (MOEA) to optimize the jitter and power of PLLs, specifically focusing on two popular fractional-N PLL architectures: digital-to-time converter (DTC)-based and harmonic-mixer (HM)-based PLLs. By applying the MOEA, we can achieve optimal jitter and power relationships for both architectures, and the observed trends in jitter and power are explained and supported with calculations.**

***Index Terms*— Phase-locked-loop, fractional N, multi-objective algorithm, phase noise, integrated jitter.**

I. Introduction

FRACTIONAL-N phase-locked loop (PLL), as shown in Fig. 1(a), is a critical component in various systems, including wireline communications, wireless communications, and data converters. It generates a frequency at a non-integer multiple of the reference and realizes a finer frequency resolution than that in integer-N PLLs. With the advent of advanced communication systems, achieving low jitter become a critical aim for PLLs. For instance, wireline transceivers using PAM4 modulation, achieving signal rates of 112 Gb/s, require PLL jitter to be less than 100 $\mathrm{fs_{rms}}$ [1], [2]. High-speed, high-resolution analog-to-digital converters (ADCs) also demand exceptionally low PLL jitter [3], [4], sometimes in the order of sub-10 $\mathrm{fs_{rms}}$ is required. While meeting these challenging jitter requirements, the power must be as low as possible for good energy efficiency. Besides, some applications prioritize power consumption in PLLs. For example, wireless sensor systems need PLLs to minimize power consumption to extend battery life [5], usually less than 2 mW or even lower. At the same time, the jitter should also be low enough within the power requirement for a better performance. However, compared to traditional integer-N PLLs, fractional-N PLLs contain non-negligible quantization noise (Q-noise) caused by the phase error in the feedback path due to the non-ideal fractional division ratio with the delta-sigma modulator (DSM) and the multi-modulus divider (MMD). Therefore, an optimal choice for design parameters of fractional-N PLLs is essential to meet the stringent jitter and power requirements for different applications.

Recently, new architectures have been proposed to mitigate the Q-noise in traditional fractional-N PLLs. One notable structure is the digital-to-time converter (DTC)-based PLL shown in Fig. 1(b), which employs a DTC to cancel the Q-noise. The DTC in the reference path is able to generate a delay that corresponds to the phase error in the feedback path caused by the DSM and MMD, realizing a Q-noise cancellation. DTC-based fractional-N PLLs have achieved excellent jitter and power performance in recent years [6], [7], [8], [9], [10]. However, the DTC is usually sensitive to process, voltage and temperature (PVT) variations, that requires complex linearity and gain calibration such as Least-Mean-Square (LMS) algorithm, necessitating an additional calibration circuit and significant convergence time.

Another promising architecture is the harmonic-mixer (HM)-based fractional-N PLL, which utilizes sample-and-hold (S/H)-based HM to suppress Q-noise [11], [12], [13], [14], [15]. Fig. 1(c) depicts a HM-based dual feedback PLL architecture proposed in [14]. This design leverages the unity

Received 19 August 2024; revised 1 November 2024 and 20 January 2025; accepted 17 February 2025. Date of publication 10 March 2025; date of current version 30 July 2025. This work was supported in part by the Institute for AI and Beyond, The University of Tokyo; in part by Japan Society for the Promotion of Science (JSPS) KAKENHI under Grant JP21H03406 and Grant JP21J21917; in part by Japan Science and Technology Agency (JST)-CRONOS under Grant JPMJCS24K1; in part by the Murata Science and Education Foundation under Grant M24AN001; and in part by the Activities of VLSI Design and Education Center (VDEC), The University of Tokyo, in collaboration with Cadence Design Systems Inc. This article was recommended by Associate Editor C.-T. Cheng. *(Yuyang Zhu and Masaru Osada are co-first authors.) (Corresponding author: Tetsuya Iizuka.)*

Yuyang Zhu and Haoming Zhang are with the Department of Electrical Engineering and Information Systems, The University of Tokyo, Tokyo 113-0032, Japan (e-mail: zhuyy55@g.ecc.u-tokyo.ac.jp; zhm@g.ecc.u-tokyo.ac.jp).

Masaru Osada was with the Department of Electrical Engineering and Information Systems, The University of Tokyo, Tokyo 113-0032, Japan. He is now with Apple, Inc., Tokyo 106-6108, Japan (e-mail: osada@silicon.u-tokyo.ac.jp).

Tetsuya Iizuka is with the Systems Design Laboratory, School of Engineering, The University of Tokyo, Tokyo 113-0032, Japan (e-mail: iizuka@vdec.u-tokyo.ac.jp).

Digital Object Identifier 10.1109/TCSI.2025.3546983

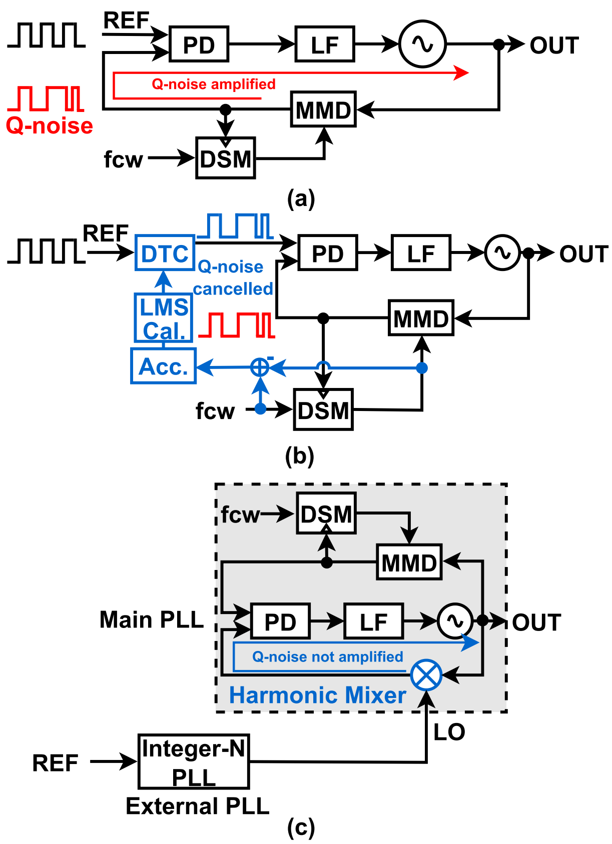


Fig. 1. Block diagrams of the (a) conventional fractional-N PLL, (b) DTC-based PLL and (c) HM-based dual-feedback PLL.

feedback gain achieved by frequency subtraction instead of the frequency division seen in the conventional architectures, avoiding amplification of the Q-noise. The HM-based architecture is calibration-free and more robust compared to the DTC-based structure because it does not use the PVT-sensitive DTC. However, the main downside of this architecture is that an additional external PLL is required to generate the local oscillation (LO) signal for the HM.

Both architectures suppress Q-noise and offer advantages over conventional single-loop fractional-N PLLs. However, DTC-based PLLs introduce extra noise and power consumption due to the DTC itself, while HM-based PLLs require an additional PLL, which also introduces extra noise and power consumption. Thus, determining which architecture is superior under different conditions necessitates optimization of these two architectures, which is a non-trivial task. Previous work [16] has addressed the jitter-power trade-off for PLLs using various approximations to achieve useful and intuitive results with reasonable calculation effort. However, this method primarily pertains to traditional integer-N PLLs. For fractional-N PLLs, the presence of Q-noise complicates matters further. Additionally, the increasing complexity of the PLL architectures introduces multiple variables, including DTC noise and power in DTC-based PLLs, and voltage-controlled oscillator (VCO) power, loop bandwidth, noise in the external PLLs of HM-based PLLs, and so on. Until now, no prior work has focused on determining the optimal jitter-power trade-off and the optimization of fractional-N PLLs, as they are not straightforward to achieve through human calculations in reasonable time due to the increased number of variables and the complexity of their relationships. Consequently, to compare different fractional-N PLLs and figure out their suitable applications given a target jitter or power consumption, a simpler and more widely-applicable optimization method is in demand.

To effectively optimize fractional-N PLLs with increasingly complex architectures, we propose a hybrid optimization approach utilizing a multi-objective evolutionary algorithm (MOEA) [17], [18], [19], [20]. This method simultaneously optimizes jitter and power of fractional-N PLLs, which can not be analyzed through calculations as done in the previous work [16]. In this paper, since DTC-based and HM-based PLLs can be applied with both LC-VCO and ring-VCO (RVCO), we primarily focus on these two cases to compare which one is better with different VCO choices. For the LC-VCO configuration, we examine the jitter-power trade-off in DTC-based PLLs and HM-based dual-feedback PLLs proposed in [14]. For the RVCO setup, HM-based dual-feedback with nested-PLL-based phase-domain low-pass filtering (PDLPF) architecture, proposed in [15], is added to the comparison. We first establish the relationship between optimal jitter and power for different fractional-N PLL architectures using the numerical method. Then detailed discussion is provided for the observed trends, underpinned by rough calculations using suitable approximations. This paper focuses on the DTC-based and HM-based architectures since they have been widely used in recent years [9], [10], [13], [15], employing different approaches to suppress the Q-noise, thereby offering a meaningful comparison. However, it is important to note that our proposed optimization approach is versatile and can be applied to other PLL designs beyond those discussed here. By utilizing MOEA, this paper presents an analytical framework that offers a more comprehensive approach to PLL optimization compared to previous works. This framework serves as a guide for designers in selecting an appropriate PLL architecture for specific applications, as well as in pursuing subsequent design steps to achieve optimal performance.

The rest of this paper is organized as follows. In Sect. II, we first introduce an MOEA that is used for optimization, and apply it to the optimization of an integer-N PLL to illustrate the feasibility of the approach. Sect. III compares different fractional-N PLL architectures with LC-VCO, including traditional fractional-N PLLs, DTC-based PLLs and HM-based dual-feedback PLLs. Sect. IV compares the cases with RVCO, and HM-based dual-feedback with nested-PLL-based PDLPF architecture is added to the comparison. Finally, we conclude our work in Sect. V.

II. Multi-Objective Optimization Algorithm

A. Introduction to MOEA

To address the conflicting objectives of jitter and power in PLL designs, we have employed an MOEA for optimization purposes. Multi-objective optimization inherently yields a set of optimal solutions, commonly known as Pareto-optimal solutions, instead of a single optimal solution. These solutions

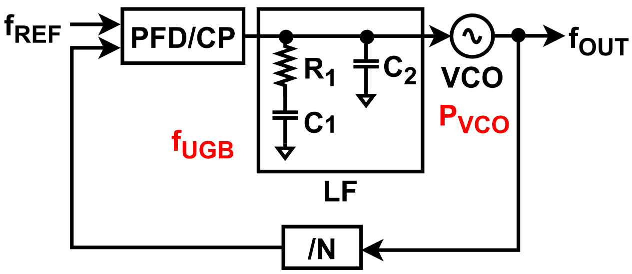


Fig. 2. A block diagram of the type-II integer-N PLL.

represent the best possible trade-offs within the defined value range, where no other solutions can simultaneously offer better performance for all objectives. In our scenario, based on the inherent trade-off between jitter and power in PLLs, the resulting Pareto-optimal solutions cater to various performance requirements. These solutions range from those that prioritize jitter to those that favor low power consumption, and others that provide a balanced compromise, thus accommodating different performance-driven applications.

In this study, the Non-dominated Sorting Genetic Algorithm II (NSGA-II) [18], [19] was selected as the MOEA. NSGA-II is an evolutionary algorithm that simulates the natural selection process characteristic of Darwinian evolution [18], which has a reduced complexity of $O(MN^2)$, where M is the number of objectives and N is the population size, rather than original NSGA [21], provides a better spread of solutions, and converges more effectively compared to other MOEAs like Pareto-Archived Evolution Strategy (PAES) [22] and Strength-Pareto EA (SPEA) [23].

In our application, the designed parameters—specifically, the power of the building blocks and the bandwidth of the PLL—serve as the chromosomes of the individuals within the population. The NSGA-II enhances the capabilities of genetic algorithm by incorporating two critical mechanisms: dominance and crowding [19]. Dominance implies that individuals with superior performance outcompete those with inferior performance, which results in a lower degree of dominance for the former. Crowding refers to the assignment of a crowding-distance value to adjacent individuals except at the boundaries. This value is calculated based on the absolute normalized difference in their function values. This strategy promotes the diversity of the Pareto-optimal solution set [20]. Each iteration of the algorithm assesses the objectives for both parent and offspring individuals, with the latter generated through the crossover and mutation of chromosomes, which are the design parameters of the PLL here. Individuals demonstrating lower dominance and higher crowding are more likely to endure through the selection process, thereby maintaining a stable population size while progressively converging towards the optimal solutions along both objective axes [20].

B. Feasibility Verification

To demonstrate the feasibility of our approach, we initially optimize a typical integer-N PLL using the NSGA-II and compared the outcomes with the calculation based on [16].

1) Analysis Framework: By following [16], a type-II third-order integer-N PLL illustrated in Fig. 2 is employed. This configuration includes a phase frequency detector (PFD), a charge pump (CP), a second-order loop filter (LF), and a voltage-controlled oscillator (VCO). In our analysis, we focus solely on the phase noise of the VCO and the reference phase noise, treating the VCO power as a representative of the total power as in [16]. We set the reference frequency, f_{REF}, to 50 MHz, and assume that the reference noise, $S_{\mathrm{n,REF}}$, is a white noise with −160 dBc/Hz level. For the VCO, we assume a simple LC-oscillator design featuring cross-coupled transistors. The phase noise spectrum of the VCO, $S_{\mathrm{n,VCO}}(f)$, is calculated based on the following equation [24]:

$$S_{\mathrm{n,VCO}}(f) = \frac{4\pi kT(1+\gamma)}{P_{\mathrm{VCO}}}\left(\frac{f_{\mathrm{OUT}}}{2Qf}\right)^2, \tag{1}$$

where k is the Boltzmann constant, T is the temperature assumed to be 300 K in this paper, γ is the excess noise coefficient of the MOSFET, assumed to be 1 in our analysis, P_{VCO} represents the power consumption of the VCO, f is the offset frequency, f_{OUT} is the output frequency of the VCO, and Q is the quality factor of the LC tank. The figure-of-merit (FoM) of the oscillator is given by [16]:

$$\mathrm{FoM} = 10\log\frac{f_{OUT}^2}{f^2 S_{\mathrm{n,VCO}}(f) P_{VCO}}. \tag{2}$$

According to (1) and (2), the FoM of the LC-VCO without considering the flicker noise is constant at all offset frequencies, and is approximated to be 186dB with $Q = 10$ in our analysis.

The algorithm has two decision variables: the open-loop unity gain bandwidth of the PLL, f_{UGB}, and the VCO power, P_{VCO}. The f_{UGB} ranges from a minimum value of 1 kHz to a maximum value of 5 MHz, which is $f_{\mathrm{REF}}/10$ due to the stability requirement of the type-II PLL with PFD/CP [25]. The VCO power varies between 100 μW and 1 W.

To proceed with the optimization, we need to express the objectives, jitter and power, in terms of these decision variables. The total power, P_{tot}, is set to equal to P_{VCO}. For the jitter, instead of using approximations to calculate, we represent the output jitter through the loop transfer function, which is more practical for actual design optimization.

To express the loop transfer function through f_{UGB}, a popular approach involves designing the LF parameters such that the ratio between the pole frequency, ω_{p}, and the unity gain bandwidth, ω_{UGB}, matches the ratio between ω_{UGB} and zero frequency, ω_{z} [26]. This ratio is denoted as a. Hence, the parameters of the LF are given by:

$$R_1 = \frac{a^2}{a^2-1}\frac{N\omega_{\mathrm{UGB}}}{K_{\mathrm{PD}}K_{\mathrm{VCO}}} \approx \frac{N\omega_{\mathrm{UGB}}}{K_{\mathrm{PD}}K_{\mathrm{VCO}}}, \tag{3}$$

$$C_1 = \frac{a^2-1}{a}\frac{K_{\mathrm{PD}}K_{\mathrm{VCO}}}{N\omega_{\mathrm{UGB}}^2} \approx \frac{aK_{\mathrm{PD}}K_{\mathrm{VCO}}}{N\omega_{\mathrm{UGB}}^2}, \tag{4}$$

$$C_2 = \frac{K_{\mathrm{PD}}K_{\mathrm{VCO}}}{aN\omega_{\mathrm{UGB}}^2}, \tag{5}$$

where $K_{\mathrm{PD}} = I_{\mathrm{p}}/2\pi$ represents the PD gain with the CP current I_{p}, and K_{VCO} denotes the VCO gain. These approximations are predicated on the assumption that $a \gg 1$. In our

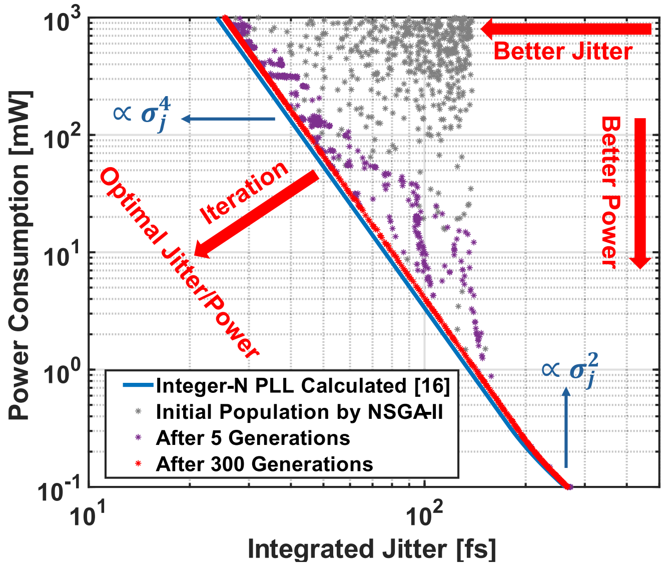


Fig. 3. Comparison between calculated results and optimization results by NSGA-II.

design, I_p is assumed to be 0.1 mA, and K_{VCO} is 100 MHz/V. The division ratio is set to 62, aligning with the values used in the subsequent sections and resulting in an output frequency, f_{OUT}, of 3.1 GHz. For the parameters in LF, we assume a to be 5, which yields a sufficient phase margin of around 67°. Then the transfer function of the LF is given by

$$H_{\mathrm{LF}}(s) = \frac{1 + sRC_1}{s(C_1 + C_2) + s^2 RC_1C_2}. \tag{6}$$

Subsequently, the open-loop transfer function is defined by

$$H_{\mathrm{OL}}(s) = \frac{K_{\mathrm{PD}}K_{\mathrm{VCO}}H_{\mathrm{LF}}(s)}{Ns}. \tag{7}$$

In H_{OL}, the only variable is the unity gain bandwidth $\omega_{\mathrm{UGB}} = 2\pi f_{\mathrm{UGB}}$.

Using these functions, we can accurately delineate the output noise contributions from each source. Specifically, the power spectral density (PSD) of the output phase noise from the reference, $S_{\mathrm{out,REF}}$, and the PSD of the output phase noise from the VCO, $S_{\mathrm{out,VCO}}$, are respectively given by

$$S_{\mathrm{out,REF}}(s) = S_{\mathrm{n,REF}}(s)\left|\frac{NH_{\mathrm{OL}}(s)}{1 + H_{\mathrm{OL}}(s)}\right|^2 \quad \text{and} \tag{8}$$

$$S_{\mathrm{out,VCO}}(s) = S_{\mathrm{n,VCO}}(s)\left|\frac{1}{1 + H_{\mathrm{OL}}(s)}\right|^2. \tag{9}$$

Finally, the total output phase noise, S_{tot} is calculated by summing these two noise:

$$S_{\mathrm{tot}}(s) = S_{\mathrm{out,REF}}(s) + S_{\mathrm{out,VCO}}(s). \tag{10}$$

Then the integrated jitter, σ_j, is derived by the integration of the S_{tot} [27] as follows:

$$\sigma_j = \frac{\sqrt{2\int_{1k}^{100M} S_{\mathrm{tot}}(s = j2\pi f)df}}{2\pi N f_{\mathrm{REF}}}. \tag{11}$$

TABLE I
THE PARAMETERS USED IN THE NSGA-II SETUP

Parameter	Detail
Population size	500
Number of generations	300
Initialization method	Random initialization within the range
Selection method	Binary tournament selection
Crossover method	Simulated Binary Crossover (SBX)
Crossover probability	0.9
Mutation method	Polynomial mutation
Mutation probability	$\frac{1}{\text{Number of Decision Variables}}$
Comparison algorithm	Crowded comparison operator
Sorting method	Non-domination-sort

In this paper, we integrate the jitter from 1 kHz to 100 MHz since the bandwidths of the PLLs are all larger than 1 kHz and smaller than 100 MHz.

With the equations and assumptions outlined above, the jitter is formulated in terms of the two decision variables, f_{UBW} and P_{VCO}. By feeding the decision variables along with their specified ranges as well as the expressions for the target objectives, σ_{j} and P_{tot}, into the NSGA-II, the algorithm autonomously initiates iterations and optimization processes. This approach enables us to efficiently explore the solution space and identify Pareto-optimal solutions that balance the trade-offs between jitter and power consumption in the PLL.

2) Optimization Results: Fig. 3 presents a comparison between the calculated results based on [16] and the optimization outcomes achieved using NSGA-II [19]. The NSGA-II is configured with a population of 500 across 300 generations, and other parameters are the same as those used in [19], which are summarized in Table I. The simulation takes less than 5 minutes in Matlab to obtain the optimization results.

In Fig. 3, the initial population, randomly generated within the predefined range, exhibits no discernible trend and demonstrates inferior jitter and power performance compared to the calculation based on [16]. However, after some number of iterations, the results move toward better performance and final optimization results after 300 generations align closely with the trend calculated in [16]. Notably, power is inversely proportional to σ_j^4 in the low-jitter region and inversely proportional to σ_j^2 in the high-jitter region. This is explained in [16] by the calculation that at low-jitter region where the optimal bandwidth is less than the maximum bandwidth ($f_{\mathrm{REF}}/10$), the presence of reference phase noise leads to $P_{\mathrm{VCO}} \propto \sigma_j^4$. However, as the optimal bandwidth can not be wider due to the maximum bandwidth limitation, the reference white noise becomes constant even as the VCO power decreases. Consequently, at low-power region, $P_{\mathrm{VCO}} \propto \sigma_j^2$. Each point in the NSGA-II results corresponds to a specific combination of decision variables, facilitating the selection of design variables applied to various application needs. The minor discrepancy between the optimization results and the calculation based on [16] is attributed to the different forms of approximation, specifically in the way of calculating the jitter. In our algorithm, we directly calculate the output jitter through the transfer function and integrate it in discrete time, while [16] employs approximations in its calculations, such as assuming that the PLL is a first-order system for simplicity of the calculation. This comparison substantiates the reliability of both the analytical calculations and the numerical optimization results.

It is important to note that regarding the jitter/power trade-off of PLLs, the primary focus is not on the exact values of the optimal jitter and power, but rather on the trends. As long as the same approximations are applied across different PLLs, and the trends of the jitter/power relationship are obtained, meaningful comparisons can be made. Therefore, the observed gap between the two sets of results is not particularly significant. As the complexity of the PLL increases, the only necessary adjustment is to modify the corresponding transfer functions. Once the output noise can be formulated, the NSGA-II algorithm is capable of finding the optimal jitter/power trade-off for the PLL.

III. Comparison Between DTC-Based and HM-Based Dual-Feedback PLL With LC-VCO

In this section, we compare the optimal jitter/power performance between DTC-based PLLs and HM-based dual feedback PLLs with LC-VCOs using NSGA-II introduced in Sect. II. For a comprehensive evaluation, traditional integer-N PLLs and fractional-N PLLs, serving as general models, are also included in the comparison.

A. Analysis Framework

1) Model Definition: In this section, we outline the framework upon which our comparative analysis is based. Fig. 4 illustrates the block diagrams and linear models of three distinct PLL architectures: a traditional fractional-N PLL, a DTC-based PLL, and a HM-based dual feedback PLL. These diagrams are essential for understanding the conditions under which the integrated jitter is calculated. Note that the nonlinearity of certain blocks, such as the PFD/CP and the DTC, can induce spurs, which affects the overall jitter. However, there is no direct trade-off between nonlinearity and power, making it challenging to formulate the impact of nonlinearity quantitatively. Therefore, as in [16], we assume all blocks to be completely linear and exclude the influence of spurs, focusing solely on the trade-off between jitter and power in this paper. For consistency in our comparison, all architectures utilize the same external reference frequency of 50 MHz and are configured to generate an identical output frequency of 3.103 GHz.

For the traditional fractional-N PLL depicted in Fig. 4(a), $\Phi_{n,\mathrm{EQUIV}}$ represents the input equivalent noise, which includes reference noise, $\Phi_{n,\mathrm{REF}}$, and PD noise, $\Phi_{n,\mathrm{PD}}$. Additionally, $\Phi_{n,\mathrm{DSM}}$ accounts for the Q-noise introduced by the DSM, and $\Phi_{n,\mathrm{VCO}}$ represents the VCO noise. In this optimization, 3 decision variables are taken into account — the PD power,

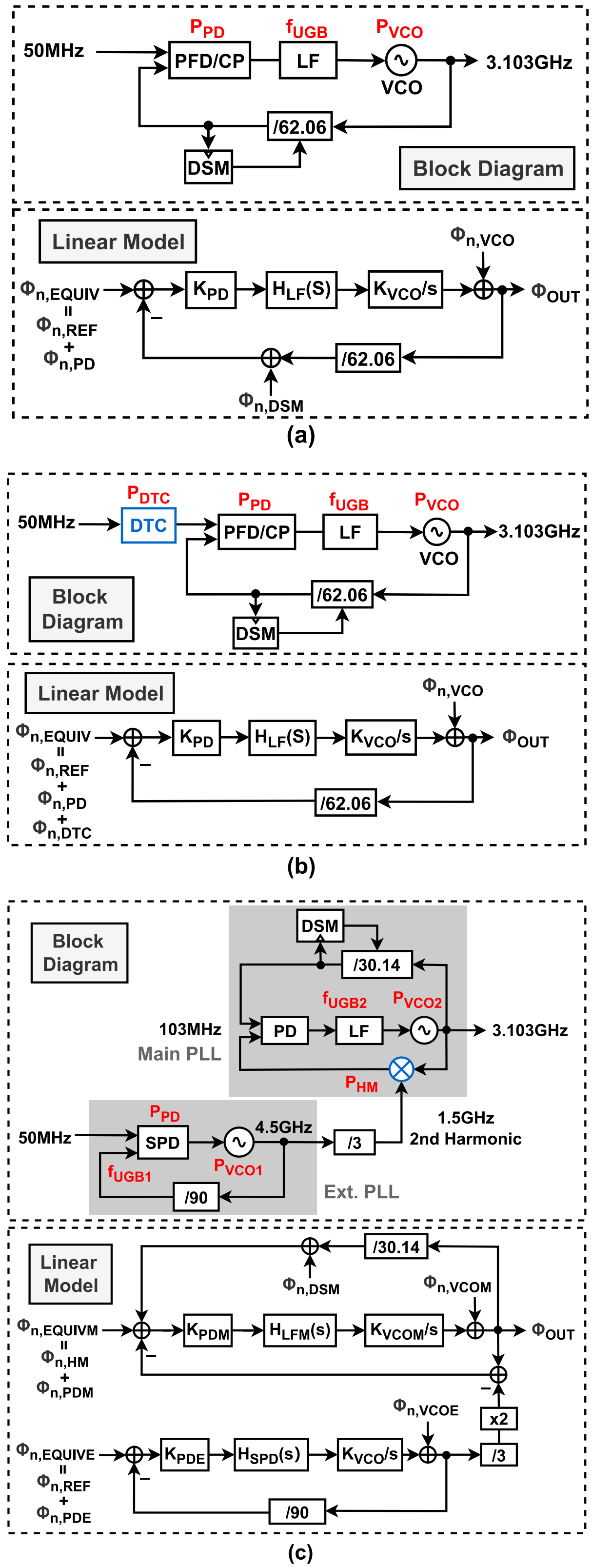


Fig. 4. Block diagrams and linear models of (a) a traditional fractional-N PLL, (b) an DTC-based PLL, and (c) a HM-based dual feedback PLL.

P_{PD}, the VCO power, P_{VCO}, and the loop bandwidth f_{UGB}, that are all marked with red in Fig. 4(a).

In the DTC-based fractional-N PLL shown in Fig. 4(b), we assume a 2$^{\mathrm{nd}}$-order DSM, along with an RC-delay-based DTC, which is widely used in previous works [8], [9], [28], for the DTC-based structure. Since the DTC with a 2[nd]-order DSM needs to provide a larger linear delay range (DR) than $2T_{\text{VCO}}$ [29], where T_{VCO} is the VCO period, for Q-noise cancellation, we assume the total DR of the DTC to be $4T_{\text{VCO}}$ as in [8]. In addition, the resolution of the DTC is assumed to be fine enough to cancel Q-noise to a negligible level, which means that the Q-noise will be completely excluded from the DSM. According to [8], [9], and [28], with an INL around 1 LSB of the DTC, the in-band fractional spur level is around −65dBc in the DTC-based fractional-N PLL, which is not significant and is comparable to the fractional spurs in the HM-based PLLs in [13], [14]. Therefore, under the assumption of the DTC-based PLL with the 2nd-order DSM, the spur caused by the DTC nonlinearity has a negligible impact on our analysis framework and can be ignored in the following discussion. However, this configuration accounts for the thermal noise of the DTC, $\Phi_{n,\text{DTC}}$, which is included in the input equivalent noise. For the decision variables, besides the 3 variables in the traditional fractional-N PLL, the DTC power, P_{DTC}, is also considered as one of the variables.

For the HM-based dual-feedback PLL in Fig. 4(c), the system comprises an external PLL and a main PLL. $\Phi_{n,\text{EQUIVE}}$ represents the input equivalent noise of the external PLL, incorporating the external PD noise $\Phi_{n,\text{PDE}}$ and the reference noise $\Phi_{n,\text{REF}}$. The external integer-N PLL generates a frequency of 4.5 GHz, which is divided by 3 to provide the LO for the HM, subsequently down-converting the overall output to 103 MHz through a 2[nd]-order harmonic mixing process for the input of the main PLL. Here, since the fine frequency tuning range at the input of the main PLL must cover the coarse frequency tunning step at the LO input of the HM [30], we assume a 4.5 GHz external PLL with a divide-by-3 for a lower frequency covering range requirement of 33.3 MHz at the main PLL input, instead of using a 1.5 GHz external PLL which requires 100 MHz frequency covering range. In the main PLL, $\Phi_{n,\text{EQUIVM}}$ denotes the input equivalent noise, which includes main PD noise $\Phi_{n,\text{PDM}}$ and HM noise $\Phi_{n,\text{HM}}$. The external VCO noise and main VCO noise are expressed as $\Phi_{n,\text{VCOE}}$ and $\Phi_{n,\text{VCOM}}$, respectively. 6 decision variables are considered for this architecture — the bandwidths of the external and main PLLs, f_{UGB1} and f_{UGB2}, the powers of the external and main VCOs, P_{VCO1} and P_{VCO2}, the power of the external PD, P_{PD} and the power of the HM, P_{HM}. It is crucial to recognize that although the dual-feedback PLL architecture includes two PDs, the total noise contribution from both is roughly equivalent to that from a single PD in other architectures. This is because the noise from the PD in the main PLL is not amplified [14] and remains negligible relative to the noise from that in the external PLL.

It is worth noting that PLLs besides the external PLL in the HM-based dual-feedback architecture are based on a type-II 3rd-order architecture [8], [14], with the zero positioned at a frequency 5× smaller than the unity gain bandwidth, while the pole is positioned at a frequency 5× greater than the unity gain bandwith. Such a design is advantageous when employing an LC-VCO, as the optimal bandwidth typically narrows, remaining significantly below the reference frequency of 50 MH. Consequently, a type-II architecture is preferred to achieve a higher loop gain. For the external PLL in the dual-feedback PLL, a type-I architecture with sampling phase detector (SPD) is employed as in [14]. Accordingly, we take into account the inherent sinc characteristic induced by the sample-and-hold operation of the SPD [31], given by

$$H_{\text{SPD}}(s = j\omega) = \frac{1}{1 + \frac{C_2}{C_1 f_{\text{CK}}} j\omega} e^{-j\pi f T_{\text{CK}}} \frac{\sin \pi f T_{\text{CK}}}{\pi f T_{\text{CK}}}, \quad (12)$$

where C_1 is usually more than 10× larger than C_2 to minimize the effect of charge sharing. $f_{CK} = T_{\text{CK}}^{-1}$ represents the clocking frequency of the SPD, which is equal to the reference frequency.

2) Noise Definition: As delineated in Sect. II, the phase noise for a 50 MHz reference is set at −160 dBc/Hz, with the VCO phase noise derived from (1). For LC-VCO, the quality factor, Q, is set to 10.

For the DSM, we assume a 2[nd]-order DSM for the DTC-based architecture, and a 3[rd]-order DSM with a MASH 1-1-1 structure in all other fractional-N PLLs. According to [32], the PSD of the DSM noise $\Phi_{n,\text{DSM}}$ is quantified using the following expression:

$$S_{\text{n,DSM}}(f) = \frac{1}{12 f_{DSM}} \left(\frac{2\pi}{N+\alpha}\right)^2 \left(2\sin\frac{\pi f}{f_{DSM}}\right)^{2(l-1)}, \quad (13)$$

where $N + \alpha$ represents the fractional division ratio of the MMD and l is the order of the DSM. f_{DSM} is the DSM clocking frequency, which is set to 50 MH in the traditional fractional-N PLL and 103 MHz in the HM-based dual-feedback PLL.

In our analysis, the white thermal noise from the PD referred to the input of the PLL is modeled as being inversely proportional to the power consumption of the PD. According to previous works [9], [10], [13], [15], no special PD designs have been applied both in the DTC-based and HM-based architectures. Therefore, for all architectures under consideration, we simply assumed that the PD contains a wide enough operation range to enable the PLL to work properly, and it contributes white noise at the reference input, inversely proportional to its power consumption, with the noise level set at −170 dBc/Hz for the PD power consumption of 1 mW. These approximations for noise levels are based on prior arts such as [8], [13], and [31]. Therefore, the noise contribution from the PDs can be calculated using the following relationship:

$$\Phi_{\text{n,EQUIVE}} = \frac{10^{-20}}{P_{\text{PD}}}. \quad (14)$$

The white thermal noise generated by the DTC is also inversely proportional to the power consumption of the DTC, P_{DTC}. Estimated from [28], we assume that the DTC operating at 1mW power consumption yields a phase noise level of −160 dBc/Hz when referred to a 50 MHz reference. Thus, the DTC noise can be expressed by

$$\Phi_{\text{n,DTC}} = \frac{10^{-19}}{P_{\text{DTC}}}. \quad (15)$$

It is important to note that the DTC output might include some residual Q-noise if the trade-off between gain calibration accuracy and reduced settling time is considered [8]. However,

TABLE II
RANGE SETTING OF DECISION VARIABLES IN THE PLL ARCHITECTURES WITH LC-VCOS

	Traditional PLL			DTC-based PLL				HM-based Dual-feedback PLL						
Var.	f_{UGB}	P_{PD}	P_{VCO}	f_{UGB}	P_{DTC}	P_{PD}	P_{VCO}	f_{UGB1}	f_{UGB2}	P_{PD}	P_{HM}	P_{VCO1}	P_{VCO2}
Min.	1 kHz	10 μW	0.1 mW	1 kHz	10 μW	10 μW	0.1 mW	1 kHz	1 kHz	10 μW	10 μW	0.1 mW	0.1 mW
Max.	5 MHz	0.1 W	1 W	5 MHz	0.1 W	0.1 W	1 W	20 MHz	10 MHz	0.1 W	0.1 W	1 W	1 W

as this analysis focuses on comparing the jitter/power trade-offs of different PLL architectures and does not consider the settling time, we will assume that the DTC calibration achieves perfect accuracy, and thus, no residual Q-noise will be considered.

Finally, the HM also adds white thermal noise at the input of the main PLL, meaning that it is not amplified. For our analysis, we assume that this noise level is -140dBc/Hz when the HM operates at a power consumption of 1mW [15], which means that the phase noise of the HM is described by the following equation:

$$\Phi_{\mathrm{n,HM}} = \frac{10^{-17}}{P_{\mathrm{HM}}}. \quad (16)$$

It is important to note that the specific values of the assumed noise are not critical in this paper, as long as they are reasonable and the relationship with power is correct. The focus of this paper is on using the algorithm to clarify the optimal jitter/power trade-off for different fractional-N PLLs, which can serve as a guide for selecting the most suitable PLL architecture for a target application and subsequent design. The phase noise values can be adjusted accordingly depending on the actual circumstances, such as different processes.

It is also worth noting note that in practical PLL designs, digital components such as the DSM can consume a non-negligible amount of power, particularly when operating at high frequencies. Notably, the power consumption of digital logic blocks typically scales with the semiconductor process technology, and there is no direct trade-off between power consumption and noise performance in these digital blocks. Therefore, for the purpose of this analysis, which aims to compare the fundamental trend of jitter/power trade-offs of different PLL architectures, we have chosen to omit the power consumed by digital logic, although this aspect could be included to enhance the accuracy of the results depending on the process being used.

B. Jitter-Power Performance Comparison

Based on the settings and methodology previously described, we explore the relationship between optimal jitter and power for the mentioned PLL architectures with LC-VCOs — traditional integer-N PLLs, traditional fractional-N PLLs, DTC-based PLLs, and HM-based dual-feedback PLLs. Based on the NSGA-II algorithm, the optimization is configured with a population size of 500 across 300 generations.

Table II summarizes the range setting of the decision variables in the above PLL structures. The bandwidths for all the type-II PLLs are varied within a range from a minimum of 1 kHz to a maximum of 10% of the reference frequency, i.e. 5 MHz for single-loop architectures with a 50 MHz

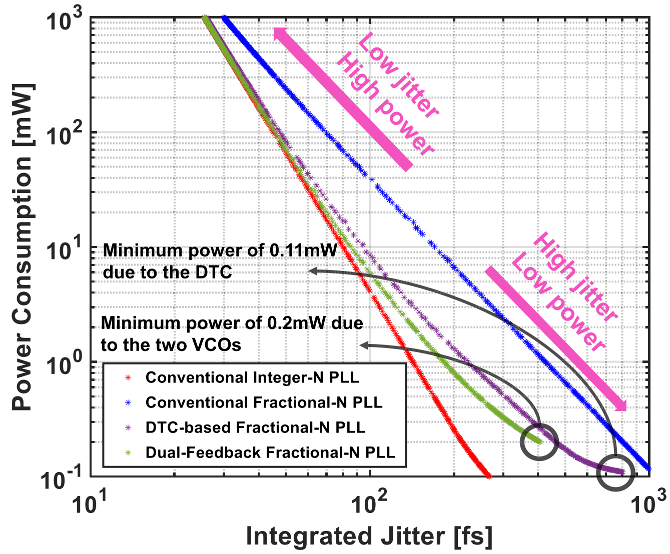


Fig. 5. The jitter/power optimization results of different PLL architectures with only VCO and DTC powers.

reference, and 10 MHz for the main PLL in the dual-feedback architecture, which operates at around a 100 MHz reference frequency. For the type-I PLL, the bandwidth is varied from 1 kHz to a maximum of 40% of the reference frequency, which is 20 MHz for the external PLL in the dual-feedback architecture here. The power consumption parameters for the VCOs are varied between 100 μW to 1 W. Meanwhile, the powers for the DTC, PD, and HM are adjusted between 10 μW and 100 mW. The power consumption range for the DTC, PD, and HM are set one-order smaller than that of the VCOs since their noise contribution tends to be much smaller. We first consider only the noise from the reference, DTC and VCO, and the power from DTC and VCO to have an idea of the general trend. Subsequently, power consumption from the PD and HM will also be included to obtain more accurate results.

1) Analysis With Only the Impacts of Reference, VCO and DTC: Fig. 5 illustrates the results obtained from our optimization analysis. The graph delineates the relationship between jitter and power across different PLL architectures considering only the impact of reference, VCO and DTC. Figs. 6(a), (b), (c), and (d) shows the phase noise plots with the integer-N PLL, the conventional fractional-N PLL, the DTC-based PLL and the HM-based PLLs, respectively, from some points in the optimization results in both low-jitter high-power and high-jitter low-power regions. Several insightful observations can be drawn from these two figures.

Firstly, for the conventional fractional-N PLL, it can be observed from Fig. 5 with blue color that the optimal jitter/power relationship appears to be almost a straight line, with the trend lying between the power being proportional

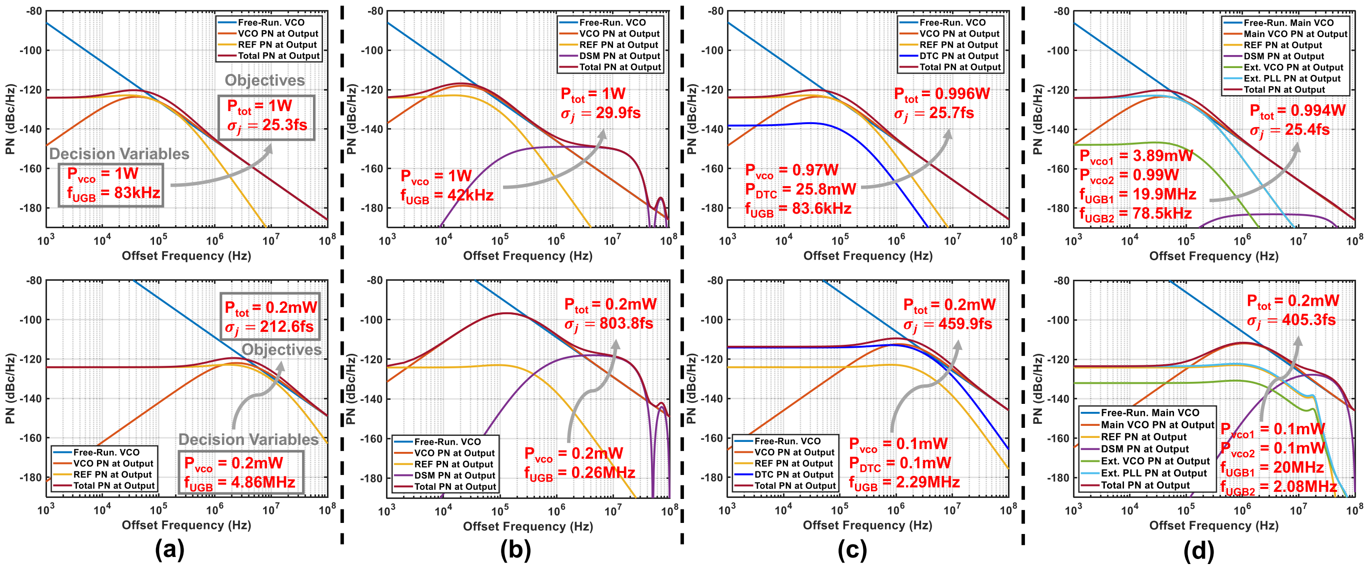


Fig. 6. The phase noise plots in low-jitter high-power regions (upper) and high-jitter low power regions (lower) of (a) the integer-N PLL, (b) the conventional fractional-N PLL, (c) the DTC-based PLL, and (d) the HM-based PLL.

to σ_j^{-3} and σ_j^{-2}. An explanation for this trend, supported by calculations, will be provided in the following paragraphs.

From the optimization result shown in Fig. 6(b), the optimized bandwidth is always much narrower than the DSM noise-shaping frequency ($f_{\mathrm{DSM}} = 50$ MHz) even when the power of the VCO is very low. Then a rudimentary explanation for this observation can be proposed by considering that the Q-noise contributes a flat noise spectrum extending from the unity gain bandwidth of the PLL, f_{BW} up to half the noise shaping frequency, as depicted in Fig. 7. Here, it is postulated that the Q-noise is approximately proportional to $f^{2(l-1)}$ where l denotes the order of the DSM. Given that the DSM is of the 3rd order, the Q-noise behavior is proportional to f^4. This Q-noise is then low-pass filtered by the characteristics of the PLL. Since the PLL is a type-II 3rd-order system, its response can be roughly approximated as flat between f_{BW} and $f_{\mathrm{DSM}}/2$, and then being proportional to $1/f^4$ at higher frequencies.

As a result, the phase noise contribution from the Q-noise manifests as a flat spectrum extending from f_{BW} to half of the DSM clocking frequency. The magnitude of this flat portion is proportional to f_{BW}^4 since the Q-noise rises proportionally to f^4 up until it reaches f_{BW} at which point it becomes flat. Finally, assuming that f_{BW} is sufficiently smaller than the half of the DSM clocking frequency, the frequency range covered by this flat noise spectrum remains relatively unaffected by variations in f_{BW}, rendering the phase noise contribution from the Q-noise at the overall PLL output proportional to f_{BW}^4. On the other hand, the phase noise contribution from the VCO is inversely proportional to the product of $P_{\mathrm{VCO}} \times f_{\mathrm{BW}}$ where P_{VCO} is the VCO power [16]. Thus, assuming that the output phase noise of the PLL is predominantly influenced by the VCO phase noise and the Q-noise, and neglecting the reference-referred equivalent white noise, we can say that

$$\sigma_j^2 \propto \alpha f_{\mathrm{BW}}^4 + \frac{\beta}{P_{\mathrm{VCO}} f_{\mathrm{BW}}}, \tag{17}$$

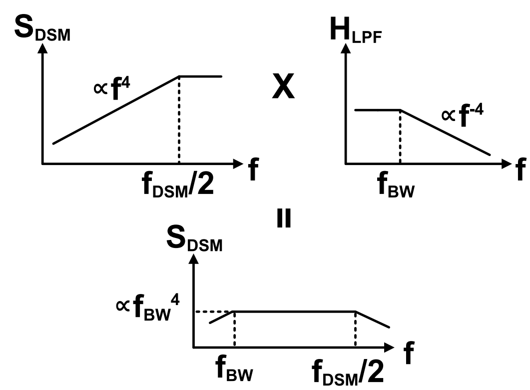


Fig. 7. A rough approximation of the DSM phase noise contribution.

where α and β are appropriately chosen constants. Based on the AM-GM inequality [33], we can transform this equation as

$$\begin{aligned} \sigma_j^2 &\propto \alpha f_{\mathrm{BW}}^4 + \frac{\beta}{P_{\mathrm{VCO}} f_{\mathrm{BW}}} \\ &\geq 5 \sqrt[5]{\frac{\alpha \beta^4}{256 P_{\mathrm{VCO}}^4}} \propto P_{\mathrm{VCO}}^{-4/5}. \end{aligned} \tag{18}$$

Therefore, if we just consider the power consumption from the VCO, the PLL power consumption is roughly proportional to $\sigma_j^{-2.5}$ in the optimally designed case. This matches the result in Fig. 5 that the power is proportional to a value between σ_j^{-3} and σ_j^{-2}.

Secondly, Fig. 5 reveals that both the DTC-based PLL (purple) and the HM-based fractional-N PLL (green) demonstrate clearly better achievable performance than the conventional fractional-N PLL over all jitter and power requirements within the analyzed regions. This improvement proves the value of the additional complexity of using a DTC or an extra PLL, for Q-noise suppression. We can also see that the dual-feedback PLL seems to offer better achievable performance than that of the DTC-based fractional-N PLL in the analyzed region, although this difference is not significant and may change depending on the practical settings and conditions.

Furthermore, it is obvious that in Fig. 5, both the DTC-based and the HM-based dual-feedback fractional-N PLLs perform worse than the conventional integer-N PLL. However, this performance difference diminishes in configurations characterized by higher power consumption and lower jitter designs.

In the case of the DTC-based fractional-N PLL, as total power consumption escalates, the power dedicated to the DTC within the optimal design framework increases until the noise generated by the DTC diminishes below the fixed reference-referred equivalent phase noise of −160 dBc/Hz. Beyond this point, as shown in the upper part of Fig. 6(c), both the power consumption and phase noise contributed by the DTC become negligible compared to that of the overall PLL, and the performance becomes similar to that of an integer-N PLL in the upper part of Fig. 6(a). In contrast, under high-jitter, low-power conditions in the lower part of Fig. 6(c), reduced power consumption significantly affects the DTC to have a higher noise than the reference white noise, making it the principal contributor to the overall phase noise along with the VCO that degrades the performance.

For the dual-feedback PLL, this trend can be attributed to the increase in total power consumption, which allows for a corresponding increase in the power allocated to the main VCO. Consequently, the optimal bandwidth for the main PLL narrows, effectively suppressing the Q-noise of DSM to negligible levels as shown in the upper graph in Fig. 6(d). Additionally, noise contributions from the external VCO, are also minimized because of the wide bandwidth external PLL and the low-pass filtering effects by the main PLL. Nevertheless, the behavior diverges at high-jitter low-power points. We can see the phase noise plot in the the lower graph of Fig. 6(d) that at high-jitter low-power regions, the bandwidth of the external PLL is inherently limited to 20 MHz by the reference frequency, and due to the higher phase noise of the main VCO, the optimal bandwidth of the main PLL also becomes wider. Therefore, the external VCO contributes more to the phase noise compared to the high-power low-jitter case as it benefits less from filtering by the main PLL. Moreover, due to the wider optimal bandwidth of the main PLL, the Q-noise from the DSM also becomes pronounced compared to the high-power low-jitter plot in the upper graph of Fig. 6(d). These factors lead to a degradation in the performance of the HM-based PLL.

For the conventional fractional-N PLL, seeing the trend in Fig. 5 and comparing the phase noise plot in the upper part of Figs. 6(a) and (b), a similar convergence in jitter and power performance of integer-N PLL is theoretically possible for the conventional fractional-N PLL at low-jitter high-power regions. This would occur when the optimal bandwidth is sufficiently narrowed to render the Q-noise insignificant. However, such conditions are not achieved within our analysis, which extends to a maximum VCO power consumption of 1W. At high-jitter low-power regions in the lower part of Fig. 6(b), the Q-noise is still pronounced in the optimal design,

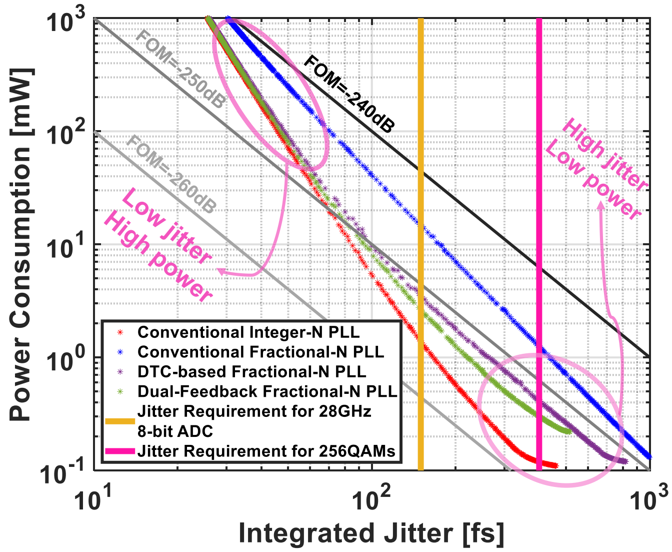


Fig. 8. The jitter/power optimization results of different PLL architectures after taking PDs and HMs into consideration.

which degrades the performance of the conventional fractional-N PLL.

Finally, in Fig. 5, it is noteworthy that the dual-feedback PLL exhibits slightly superior performance to the DTC-based PLL in medium jitter and power regions. However, this advantage gradually diminishes in the design of extremely low-power and high-jitter regions. The initial benefit in low to medium jitter scenarios stems from the ability to suppress the overhead noise introduced by the additional VCO in the dual-feedback PLL through the use of a wide bandwidth for external PLL and narrow bandwidth for the main PLL, which effectively minimizes the noise contribution from the external PLL. Nevertheless, as the design constraints shift towards lower power, the power consumption of the VCOs reaches its practical minimum limit. Consequently, for HM-based PLL, the optimal bandwidth of the main PLL is wider, reducing its ability to filter the noise from the external PLL and DSM, thus, leading to a degradation in overall performance.

2) Analysis Taking PDs and HMs Into Consideration: So far, our analysis has focused solely on the power consumption contributions from the VCO and the DTC. To enhance the accuracy of our comparison, we will now also consider the power and noise from the PDs, as well as from the HM. Fig. 8 shows the results obtained. A contour plot for the jitter-power FoM with black and grey, alongside the vertical lines representing the jitter requirements for specific applications are included. The yellow line indicates a jitter threshold of 150 fs, required for a clock of a 28 GHz ADC to achieve an Effective Number of Bits (ENOB) of 8, a common specification for ADCs utilized in 56 Gb/s or 112 Gb/s wireline receivers [34], [35], [36], [37]. The pink line indicates a jitter requirement of 400 fs, necessary for the 256 QAM transceiver to meet the −43 dBc integrated phase noise (IPN) requirement around 3 GHz [38], [39].

At low-jitter, high-power design points in Fig. 8, all the trends are very similar to the one in Fig. 5, primarily because the power overheads from the reference path and the HM

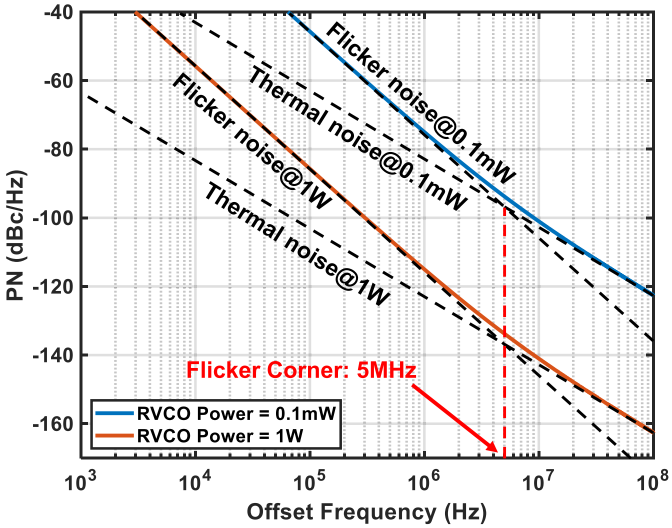


Fig. 9. The phase noise plot of the RVCO at 3.103 GHz.

are negligible. However, at high-jitter and low-power design points, power consumption tends to saturate due to the minimum power requirements of the various blocks, and the noise from the PD becomes non-negligible relative to the fixed reference noise. Despite these factors, the performance of both the DTC-based PLL and the dual-feedback PLL remains superior to that of the conventional fractional-N PLL across a wide range of conditions. Importantly, the performance advantage of these PLLs is most pronounced in the jitter range of approximately 100 to 200 fs, which aligns with crucial application requirements. Furthermore, the best achievable FoM for the dual-feedback PLL is roughly equivalent to that of the DTC-based PLL, surpassing that of the conventional fractional-N PLL and trailing behind that of the conventional integer-N PLL.

Based on the results in Fig. 8, we have analytically demonstrated, through a hybrid approach combining numerical methods, that the dual-feedback architecture can achieve better performance than the conventional fractional-N PLL and comparable performance to the DTC-based fractional-N PLL, depending on specific settings and targeted jitter/power metrics. In practice, the use of multiple VCOs in the HM-based architecture [11], [12], [14] requires an additional area, which could be a potential disadvantage compared to the DTC-based architectures [6], [7], [8], [9], [10]. However, the HM-based dual-feedback architecture offers a distinct advantage, i.e. no calibration required unlike the DTC-based architectures, which results in a shorter settling time and a simpler overall design.

IV. Jitter-Power Performance Comparison With Ring-VCO

In advanced wireless and wireline transceivers, especially for multiple-input and multiple-output (MIMO) systems [9], [40], [41] and transceivers for high-speed links [42] where multiple PLLs are required, RVCO-based PLLs are increasingly favored over their LC counterparts, since they usually occupy a smaller area. Therefore, enhancing the performance of RVCO-based fractional-N PLLs is essential for such applications.

The DTC-based structure, known for effectively suppressing Q-noise, can be easily integrated with an RVCO. Similarly, the HM-based architecture, which prevents the amplification of the Q-noise, is also compatible with an RVCO. Furthermore, considering the worse noise performance of RVCOs compared to LC-VCOs, additional techniques such as PDLPF [15], [43], [44] are usually employed to filter Q-noise for a wide PLL bandwidth to suppress the noise from RVCOs. Consequently, as proposed in [15], a nested-PLL serving as a PDLPF with a split-feedback divider is added to the HM-based dual-feedback structure to further filter Q-noise, and this configuration is included in our comparison with RVCOs. However, for LC-VCOs, the optimal PLL bandwidth is usually narrower due to the lower VCO noise, making additional filtering by the nested-PLL unnecessary in the HM-based dual-feedback structure, which will bring additional design complexity and increased power consumption. Therefore, we include the structure with PDLPF only in the RVCO case.

So far, we have shown that with LC-VCO at specific output frequencies, HM-based PLLs can slightly outperform DTC-based PLLs in optimal design configurations, and both exhibit superior performance compared to the traditional fractional-N PLLs. However, a few questions remain: Is this performance advantage maintained with RVCO? Is it really worthy to incorporate an additional PLL as a phase-domain noise filter in the HM-based PLL architecture? To address these questions, this section will undertake a similar optimization approach as presented in Sect. III, evaluating three structures: DTC-based, HM-based dual-feedback, and HM-based dual-feedback with PDLPF. We will compare their performance within a targeted range to determine which architecture offers the best performance for RVCO-based systems.

A. Analysis Framework

Considering the worse phase noise performance of the RVCOs compared with the LC counterparts, some of the analysis settings need to be adjusted. In the LC-VCO case, flicker noise is ignored as in [13] and [16], since it is typically low enough to have a negligible impact on the analysis results. However, for the RVCOs, in addition to increased white noise, the phase noise is largely influenced by the flicker noise at low offset frequencies, with a slope of −30 dB/dec. Therefore, the impact of the flicker noise cannot be ignored and is taken into account in the RVCO cases. According to [45], the phase noise of an RVCO can be divided into two components:

$$S_{\mathrm{n,RVCO}}(f) = S_{\mathrm{white,RVCO}}(f) + S_{\mathrm{flicker,RVCO}}(f), \tag{19}$$

where $S_{\mathrm{white,RVCO}}(f)$ represents the RVCO phase noise due to the the white noise, which can be calculated using (1) with $Q = 1$, and $S_{\mathrm{flicker,RVCO}}(f)$ represents the RVCO phase noise due to the flicker noise, which is given by

$$S_{\mathrm{flicker,RVCO}}(f) = \frac{K_{\mathrm{f}}}{P_{\mathrm{VCO}}} \frac{f_{\mathrm{OUT}}^2}{f^3}, \tag{20}$$

where K_{f} is a constant depending on the factors such as power supply voltage, the number of stages, and the device excess

TABLE III
RANGE SETTING OF DECISION VARIABLES IN THE PLL ARCHITECTURES WITH RING-VCOS

	Traditional PLL			DTC-based PLL				HM-based Dual-feedback PLL						HM-based Dual-feedback PLL with Split Divider and PDLPF							
Var.	f_{UGB}	P_{PD}	P_{VCO}	f_{UGB}	P_{DTC}	P_{PD}	P_{VCO}	f_{UGB1}	f_{UGB2}	P_{PD}	P_{HM}	P_{VCO1}	P_{VCO2}	f_{UGB1}	f_{UGB2}	f_{UGB3}	P_{PD}	P_{HM}	P_{VCO1}	P_{VCO2}	P_{VCO3}
Min.	1kHz	10μW	0.1mW	1kHz	10μW	10μW	0.1mW	1kHz	1kHz	10μW	10μW	0.1mW	0.1mW	1kHz	1kHz	1kHz	10μW	10μW	0.1mW	0.1mW	0.1mW
Max.	5MHz	0.1W	1W	10MHz	0.1W	0.1W	1W	10MHz	10MHz	0.1W	0.1W	1W	1W	20MHz	10MHz	60MHz	0.1W	0.1W	1W	1W	1W

noise coefficient. The flicker corner is defined as the frequency at which $S_{\text{white,RVCO}}(f) = S_{\text{flicker,RVCO}}(f)$. According to previous studies [15], [46], [47], the flicker corner typically lies between 1 MHz and 10 MHz for RVCOs below 5 GHz. Therefore, to account for the flicker noise of the RVCO, we assume that the flicker corner of the RVCO is fixed at an offset frequency of 5 MHz. Under this assumption, K_f can be calculated as 1.3×10^{-13}, allowing us to determine the RVCO phase noise using (19).

Fig. 9 shows the phase noise plot of the RVCO under different powers at the PLL output frequency of 3.103 GHz calculated from (19) and (20). The FoMs of the RVCO at different offset frequencies, calculated by (2), are 158.1 dB, 164.1 dB and 165.6 dB at 1 MHz, 10 MHz and 100 MHz, respectively.

For RVCO-based PLLs, a wide bandwidth is typically used to suppress the inherently high noise levels associated with RVCOs, which is often achieved using an SPD with a high gain [31]. To suppress flicker noise at low frequency offsets, a dual-path structure is usually employed for integer-N and DTC-based fractional-N PLLs [9], [39], [48], [49]. Consequently, following the structures from previous studies [9], [39], [48], [49], we assume a type-II SPD-based dual-path PLL for the architectures of the integer-N and DTC-based PLLs, where the unity gain bandwidth ranges from 1 kHz to 10 MHz, which is 20% of the reference frequency. The pole introduced by the integral path is placed at a frequency 5× smaller than the bandwidth, resulting in a reasonable phase margin of approximately 45° at the maximum bandwidth. The traditional fractional-N PLL continues to use a type-II PLL structure with PFD/CP, with bandwidth varies from 1 kHz to 5 MHz. This is because, even with an RVCO, the Q-noise in traditional fractional-N PLLs remains substantial, necessitating a narrower optimal bandwidth less than 10% of the reference frequency. Furthermore, the nonlinearity of the SPD [31] makes it unsuitable for architectures where Q-noise is not canceled beforehand. The HM-based dual-feedback PLL follows the configurations described in [14], as previously discussed in Sect. III, while the HM-based dual-feedback PLL with PDLPF adopts the structure presented in [15]. This includes a type-I PLL with an SPD for the external PLL and a type-II PLL with a PFD/CP for the main PLL and nested PLL. For the split-feedback divider, the first and the second stage dividers have division ratios of 5.02 and 6, respectively, thereby increasing the noise shaping frequency to 618 MH. The bandwidth of the nested PLL is varied from 1 kHz to 60 MHz. All other settings remain consistent with those employed in Sect. III. The range settings for the decision variables in the PLL architectures with RVCOs are summarized in Table III.

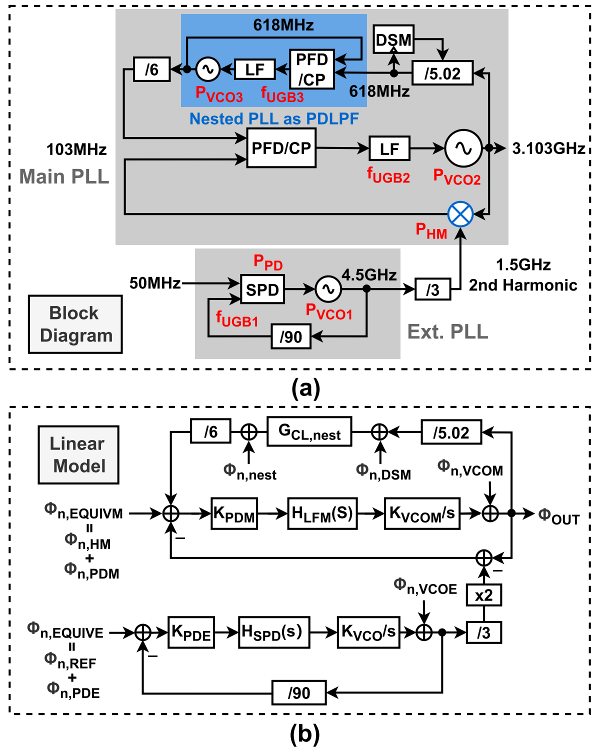


Fig. 10. (a) Block diagram and (b) linear model of the HM-based dual-feedback PLL with PDLPF.

Fig. 10 shows the block diagram and linear model of the HM-based dual-feedback PLL with PDLPF. The nested PLL in the main PLL is a traditional type-II integer-N PLL, same as the one in Sect. II, with a unity division ratio. In Fig. 10(b), $G_{CL,nest}$ represents the closed-loop transfer function of the nested PLL, and $\Phi_{n,nest}$ is the output equivalent noise from the nested PLL. Here, in the nested PLL, only the VCO noise is considered since the noise from the PFD/CP is not amplified.

B. Jitter-Power Performance Comparison

The optimization results are summarized in Fig. 11. It is evident that both the DTC-based PLL (purple) and the dual-feedback PLLs (green and orange) exhibit superior performance compared to the conventional fractional-N PLL (blue), yet they fall short of the performance achieved by the conventional integer-N PLL (red). Notably, the performance of the dual-feedback PLL is enhanced when the split-feedback divider and PDLPF are employed, although this improvement is not observed at low power and high jitter design points. At these points, the mandatory minimum power consumption of the three VCOs results in degraded performance.

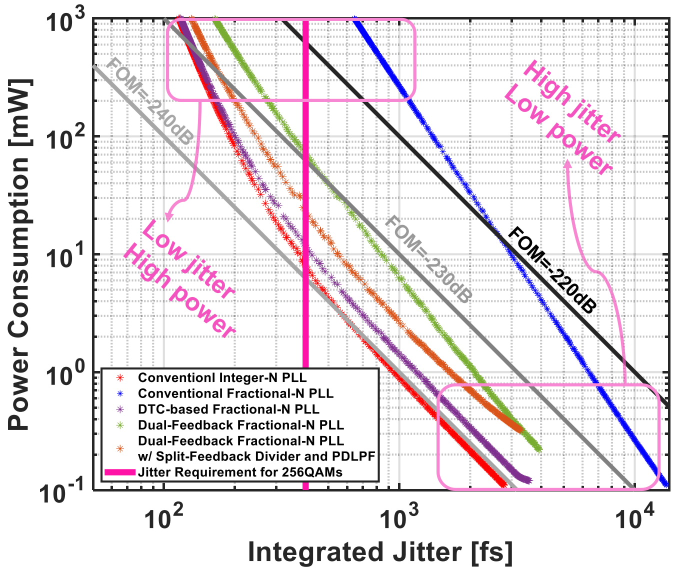


Fig. 11. The jitter/power optimization results of different PLL architectures with RVCOs.

On the other hand, both of the dual-feedback PLLs perform worse than the DTC-based fractional-N PLL in this RVCO case. However, HM-PLLs still have the additional benefit of calibration-less feature. Regarding the potential applications, RVCO-based PLLs generally perform worse than LC-VCO-based PLLs and are impractical for applications requiring low jitter, such as providing the clock for a 28 GHz ADC, because it may require excessive power consumption, even more than 100 mW. Therefore, we present only the jitter requirement for 256QAM applications at approximately 3 GHz [38], [39], as shown with a pink vertical line in Fig. 11. With RVCOs, the DTC-based PLL architecture is more preferable for 256QAM applications regarding the jitter/power performance.

It is notable that this analysis shows that the dual-feedback fractional-N PLL underperforms the DTC-based fractional-N PLL, contrasting with the findings in Sect. III, where the dual-feedback PLL with an LC-VCO demonstrated comparable or even superior performance under certain conditions. This difference primarily stems from the inherent phase noise performance differences between RVCOs and LC-VCOs at the equivalent power consumptions. With RVCOs typically exhibiting worse phase noise characteristics, the optimal PLL bandwidth tends to be wider. This wider bandwidth has a more detrimental impact on the dual-feedback PLL than on the DTC-based PLL. This effect arises because the additional noise source in the dual-feedback PLL is the shaped Q-noise, which is more sensitive to the bandwidth expansion, whereas in the DTC-based PLL, the noise is predominantly white noise from the DTC with a flat profile.

V. Conclusion

This paper employs MOEA to optimize jitter-power trade-offs in various fractional-N PLL architectures focusing on reducing Q-noise. It allows for comparisons of the optimal jitter-power trade-off between any PLL architecture and the optimization of PLL parameters without complex calculations once the linear model is established. With the comparisons, the most suitable architecture for a specific target application can be selected.

Using this method, the optimal jitter-power trends of two popular fractional-N PLL architectures — DTC-based and HM-based PLLs — with both LC-VCO and RVCOs are obtained and compared. From the comparison results, it is observed that with LC-VCOs, HM-based PLL performs slightly better than the DTC-based PLL, while with RVCOs, although the addition of PDLPF and the split feedback divider improves the performance of the HM-based PLL, the DTC-based PLL performs better than the HM-based PLLs. The presented comparison framework helps to determine the most suitable architecture for specific applications such as wireless or wireline communications. This analysis aims to elucidate distinctions in performance across different PLL architectures, thereby providing insight into their respective efficacy under various application conditions.

Based on the proposed method, several future research directions are expected. Firstly, the proposed method can be applied to optimize more PLL architectures, like sub-sampling fractional-N PLLs and injection-locking PLLs, for different applications. Secondly, more performance metrics can be formulated and evaluated in the future work, with the use of new algorithms capable of optimizing more objectives, while the algorithm currently used in this paper only supports the optimization of two objectives. Moreover, most existing research directly focuses on the relationship between jitter and power, with limited works formulating the relationships between jitter/power and other metrics such as spurs and area. Future work could therefore focus on formulating and evaluating the relationships between more design parameters, e.g. area, reference/fractional spurs and the impact from the nonlinearity of the circuit components, for the optimization of more objectives.

With more objectives and increasingly complex structures, it is important to explore different algorithms, such as Hybrid Genetic Algorithms (HGA) [50], and assess the strengths and weaknesses of different algorithms. Meanwhile, training the algorithms to prevent convergence issues or getting trapped in local optima [48], [51], will become essential. This paper lays the foundation for future research, providing a fundamental approach to assist designers in determining the design parameters of fractional-N PLLs to achieve optimal jitter and power efficiently.

References

[1] J. Im et al., "A 112 Gb/s PAM-4 long-reach wireline transceiver using a 36-way time-interleaved SAR ADC and inverter-based RX analog front-end in 7 nm FinFET," *IEEE J. Solid-State Circuits*, vol. 56, no. 1, pp. 7–18, Jan. 2021.

[2] E. Chong et al., "A 112Gb/s PAM-4, 168Gb/s PAM-8 7bit DAC-based transmitter in 7 nm FinFET," in *Proc. IEEE 47th Eur. Solid State Circuits Conf. (ESSCIRC)*, Sep. 2021, pp. 523–526.

[3] L. Kull et al., "A 24-to-72 GS/s 8b time-interleaved SAR ADC with 2.0-to-3.3pJ/conversion and >30dB SNDR at Nyquist in 14nm CMOS FinFET," in *Proc. IEEE Int. Solid - State Circuits Conf. (ISSCC)*, Feb. 2018, pp. 358–360.

[4] E. Martens et al., "A 42 GS/s 7b 16 nm massively time-interleaved slope-ADC," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2024, pp. 396–398.

[5] A. Paidimarri, N. Ickes, and A. P. Chandrakasan, "A 0.68 V 0.68 mW 2.4GHz PLL for ultra-low power RF systems," in *Proc. IEEE Radio Freq. Integr. Circuits Symp. (RFIC)*, May 2015, pp. 397–400.

[6] S. Levantino, G. Marzin, and C. Samori, "An adaptive pre-distortion technique to mitigate the DTC nonlinearity in digital PLLs," *IEEE J. Solid-State Circuits*, vol. 49, no. 8, pp. 1762–1772, Aug. 2014.

[7] J. Z. Ru, C. Palattella, P. Geraedts, E. Klumperink, and B. Nauta, "A high-linearity digital-to-time converter technique: Constant-slope charging," *IEEE J. Solid-State Circuits*, vol. 50, no. 6, pp. 1412–1423, Jun. 2015.

[8] W. Wu et al., "A 28 nm 75 fsrms analog fractional-N sampling PLL with a highly linear DTC incorporating background DTC gain calibration and reference clock duty cycle correction," *IEEE J. Solid-State Circuits*, vol. 54, no. 5, pp. 1254–1265, May 2019.

[9] H. Park, C. Hwang, T. Seong, and J. Choi, "A low-jitter ring-DCO-based fractional-N digital PLL with a 1/8 DTC-range-reduction technique using a quadruple-timing-margin phase selector," *IEEE J. Solid-State Circuits*, vol. 57, no. 12, pp. 3527–3537, Dec. 2022.

[10] M. Rossoni et al., "An 8.75 GHz fractional-N digital PLL with a reverse-concavity variable-slope DTC achieving 57.3 fs_{rms} integrated jitter and 252.4dB FoM," in *Proc. IEEE Int. Solid-State Circuits Conf. (ISSCC)*, Feb. 2024, pp. 188–190.

[11] D. Yang et al., "A calibration-free triple-loop bang-bang PLL achieving $131fs_{\mathrm{rms}}$ jitter and -70dBc fractional spurs," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2019, pp. 266–268.

[12] D. Yang, D. Murphy, H. Darabi, A. Behzad, R. Ruby, and R. Parker, "An FBAR driven -261dB FOM fractional-N PLL," in *Proc. IEEE Radio Freq. Integr. Circuits Symp. (RFIC)*, Jun. 2021, pp. 147–150.

[13] D. Yang et al., "A harmonic-mixing PLL architecture for millimeter-wave application," *IEEE J. Solid-State Circuits*, vol. 57, no. 12, pp. 3552–3566, Dec. 2022.

[14] M. Osada, Z. Xu, and T. Iizuka, "A 3.2-to-3.8 GHz harmonic-mixer-based dual-feedback fractional-N PLL achieving -65 dBc in-band fractional spur," *IEEE Solid-State Circuits Lett.*, vol. 3, pp. 534–537, 2020.

[15] M. Osada, Z. Xu, Z. Yang, and T. Iizuka, "A fractional-N ring PLL using harmonic-mixer-based dual-feedback and split-feedback frequency division with phase-domain filtering," *IEEE J. Solid-State Circuits*, vol. 59, no. 7, pp. 2171–2184, Jul. 2024.

[16] B. Razavi, "Jitter-power trade-offs in PLLs," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 68, no. 4, pp. 1381–1387, Apr. 2021.

[17] K. Deb, "Multi-objective genetic algorithms: Problem difficulties and construction of test problems," *Evol. Comput.*, vol. 7, no. 3, pp. 205–230, Sep. 1999.

[18] R. L. Haupt, "An introduction to genetic algorithms for electromagnetics," *IEEE Antennas Propag. Mag.*, vol. 37, no. 2, pp. 7–15, Apr. 1995.

[19] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, "A fast and elitist multiobjective genetic algorithm: NSGA-II," *IEEE Trans. Evol. Comput.*, vol. 6, no. 2, pp. 182–197, Apr. 2002.

[20] Z. Wu et al., "A multiobjective design architecture for versatile stretchable touchscreen displays," *IEEE Sensors J.*, vol. 23, no. 18, pp. 21956–21965, Sep. 2023.

[21] N. Srinivas and K. Deb, "Muiltiobjective optimization using nondominated sorting in genetic algorithms," *Evol. Comput.*, vol. 2, no. 3, pp. 221–248, Sep. 1994.

[22] J. Knowles and D. Corne, "The Pareto archived evolution strategy: A new baseline algorithm for Pareto multiobjective optimisation," in *Proc. IEEE Congr. Evol. Comput. (CEC)*, vol. 1, Aug. 1999, pp. 98–105.

[23] E. Zitzler, *Evolutionary Algorithms for Multiobjective Optimization: Methods and Applications*. Aachen, Germany: Shaker, 1999.

[24] P. Andreani, X. Wang, L. Vandi, and A. Fard, "A study of phase noise in colpitts and LC-tank CMOS oscillators," *IEEE J. Solid-State Circuits*, vol. 40, no. 5, pp. 1107–1118, May 2005.

[25] F. Gardner, "Charge-pump phase-lock loops," *IEEE Trans. Commun.*, vol. COM-28, no. 11, pp. 1849–1858, Nov. 1980.

[26] *PLL Loop Filter Design and Fine Tuning*, document R31AN0034EU0100, Renesas Electron. Corp., 2023.

[27] B. Razavi, *Design of CMOS Phase-Locked Loops*. Cambridge, U.K.: Cambridge Univ. Press, 2020.

[28] W. Wu et al., "A 14-nm ultra-low jitter fractional-N PLL using a DTC range reduction technique and a reconfigurable dual-core VCO," *IEEE J. Solid-State Circuits*, vol. 56, no. 12, pp. 3756–3767, Dec. 2021.

[29] T. M. Vo, C. Samori, A. L. Lacaita, and S. Levantino, "A novel segmentation scheme for DTC-based $\Delta\Sigma$ fractional-N PLL," in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, May 2017, pp. 1–4.

[30] M. Osada, Z. Xu, R. Shibata, and T. Iizuka, "Analysis of offset spurs in phase-locked-loops employing harmonic-mixer-based feedback with sample-and-hold operation," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 69, no. 12, pp. 5072–5084, Dec. 2022.

[31] L. Kong and B. Razavi, "A 2.4 GHz 4 mW integer-N inductorless RF synthesizer," *IEEE J. Solid-State Circuits*, vol. 51, no. 3, pp. 626–635, Mar. 2016.

[32] M. H. Perrott, M. D. Trott, and C. G. Sodini, "A modeling approach for Σ-Δ fractional-N frequency synthesizers allowing straightforward noise analysis," *IEEE J. Solid-State Circuits*, vol. 37, no. 8, pp. 1028–1038, Aug. 2002.

[33] D. A. Klarner, *The M Athematical Gardner*. New York, NY, USA: Springer, 2012.

[34] D. Pfaff, R. Abbott, X.-J. Wang, S. Moazzeni, R. Mason, and R. R. Smith, "A 14 GHz bang-bang digital PLL with sub-150-fs integrated jitter for wireline applications in 7 nm FinFET CMOS," *IEEE J. Solid-State Circuits*, vol. 55, no. 3, pp. 580–591, Mar. 2020.

[35] J. Kim et al., "A 112Gb/s PAM-4 transmitter with 3-tap FFE in 10 nm CMOS," in *Proc. IEEE Int. Solid State Circuits Conf. (ISSCC)*, Feb. 2018, pp. 102–104.

[36] P. Upadhyaya et al., "A fully adaptive 19-to-56Gb/s PAM-4 wireline transceiver with a configurable ADC in 16nm FinFET," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2018, pp. 108–110.

[37] M. Raj, A. Bekele, D. Turker, P. Upadhyaya, Y. Frans, and K. Chang, "A $164fs_{\mathrm{rms}}$ 9-to-18 GHz sampling phase detector based PLL with in-band noise suppression and robust frequency acquisition in 16nm FinFET," in *Proc. Symp. VLSI Circuits*, Jun. 2017, pp. C182–C183.

[38] Y. Jo, J. Kim, Y. Shin, C. Hwang, H. Park, and J. Choi, "A $135fs_{\mathrm{rms}}$-jitter 0.6-to-7.7 GHz LO generator using a single LC-VCO-based subsampling PLL and a ring-oscillator-based sub-integer-N frequency multiplier," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2023, pp. 1–3.

[39] C. Hwang, H. Park, Y. Lee, T. Seong, and J. Choi, "A low-jitter and low-fractional-spur ring-DCO-based fractional-N digital PLL using a DTC's second-/third-order nonlinearity cancellation and a probability-density-shaping $\Delta\Sigma$M," *IEEE J. Solid-State Circuits*, vol. 57, no. 9, pp. 2841–2855, Sep. 2022.

[40] H. Yoon et al., "A -31dBc integrated-phase-noise 29 GHz fractional-N frequency synthesizer supporting multiple frequency bands for backward-compatible 5G using a frequency doubler and injection-locked frequency multipliers," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2018, pp. 366–368.

[41] J. Lee et al., "6.1 a low-power and low-cost 14 nm FinFET RFIC supporting legacy cellular and 5G FR1," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, vol. 64, Feb. 2021, pp. 90–92.

[42] P. Mishra et al., "A 112Gb/s ADC-DSP-based PAM-4 transceiver for long-reach applications with >40dB channel loss in 7 nm FinFET," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, vol. 64, Feb. 2021, pp. 138–140.

[43] P. Park, D. Park, and S. Cho, "A 2.4 GHz fractional-N frequency synthesizer with high-OSR modulator and nested PLL," *IEEE J. Solid-State Circuits*, vol. 47, no. 10, pp. 2433–2443, Oct. 2012.

[44] L. Feng, W. Rhee, and Z. Wang, "A DTC-free fractional-N BBPLL with FIR-embedded injection-locked-oscillator-based phase-domain lowpass filter," *IEEE J. Solid-State Circuits*, vol. 59, no. 3, pp. 728–739, Mar. 2024.

[45] A. A. Abidi, "Phase noise and jitter in CMOS ring oscillators," *IEEE J. Solid-State Circuits*, vol. 41, no. 8, pp. 1803–1816, Aug. 2006.

[46] A. Santiccioli, M. Mercandelli, A. L. Lacaita, C. Samori, and S. Levantino, "A 1.6-to-3.0 GHz fractional-N MDLL with a digital-to-time converter range-reduction technique achieving 397-fs jitter at 2.5-mW power," *IEEE J. Solid-State Circuits*, vol. 54, no. 11, pp. 3149–3160, Nov. 2019.

[47] Z. Yang, Z. Xu, M. Osada, and T. Iizuka, "A 10 GHz inductorless cascaded PLL with zero-ISF subsampling phase detector achieving -63-dBc reference spur, 175-fs RMS jitter and -240-dB FOMjitter," in *Proc. IEEE Symp. VLSI Technol. Circuits (VLSI Technol. Circuits)*, Jun. 2022, pp. 10–11.

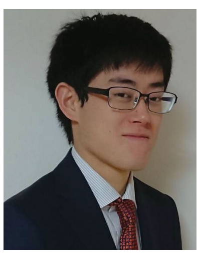
[48] Y. Zhou, Z. He, C. Chen, T. Wang, L. Xiao, and X. Wang, "An efficient power optimization approach for fixed polarity Reed–Muller logic circuits based on metaheuristic optimization algorithm," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 41, no. 12, pp. 5380–5393, Dec. 2022.

[49] Y. Huang, Y. Chen, Z. Yang, R. P. Martins, and P.-I. Mak, “A 0.027 mm^2 5.6–7.8 GHz ring-oscillator-based ping-pong sampling PLL scoring 220.3fs_{rms} jitter and -74.2dBc reference spur,” in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2024, pp. 130–132.

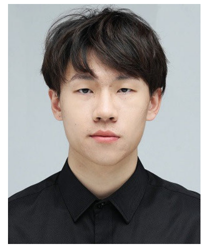
[50] L. Gharsalli, “Hybrid genetic algorithms,” in *Optimisation Algorithms and Swarm Intelligence*, N. Vakhania and M. E. Aydin, Eds., Rijeka, Croatia: IntechOpen, 2022, ch. 2, doi: 10.5772/intechopen.104735.

[51] Y. Zhou et al., “An efficient and fast area optimization approach for mixed polarity Reed–Muller logic circuits,” *Chin. J. Electron.*, vol. 33, no. 5, pp. 1165–1180, Sep. 2024.

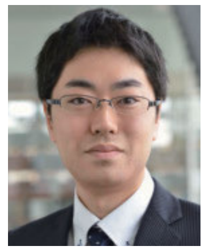
Yuyang Zhu (Graduate Student Member, IEEE) received the B.S. degree in opto-electronic information science and engineering from Sun Yat-sen University, Guangzhou, China, in 2021, and the M.S. degree in electrical engineering from The University of Tokyo, Tokyo, Japan, in 2024, where he is currently pursuing the Ph.D. degree. His current research interests include high performance inductorless frequency synthesizers for wireless and wireline communications.



Masaru Osada (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electrical engineering from The University of Tokyo, Tokyo, Japan, in 2019, 2021, and 2024, respectively. He is currently with Apple Inc., as an Analog Design Engineer. His current research interests include high-performance frequency synthesizers for wireless and wireline communications.



Haoming Zhang (Graduate Student Member, IEEE) received the B.E. degree in integrated circuit and system from Dalian University of Technology, Dalian, China, in 2021, and the M.E. degree in electrical engineering and information systems from The University of Tokyo, Tokyo, Japan, in 2023, where he is currently pursuing the Ph.D. degree with the Department of Electrical Engineering and Information Systems. His current research interests include characterization and design automation of integrated circuits and high-performance frequency synthesizers for wireless and wireline communications.



Tetsuya Iizuka (Senior Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electronic engineering from The University of Tokyo, Tokyo, Japan, in 2002, 2004, and 2007, respectively.

From 2007 to 2009, he was with THine Electronics Inc., Tokyo, as a High-Speed Serial Interface Circuit Engineer. He joined The University of Tokyo in 2009, where he is currently an Associate Professor with the Systems Design Laboratory, School of Engineering. From 2013 to 2015, he was a Visiting Scholar with the University of California at Los Angeles, CA, USA. His current research interests include data conversion techniques, high-speed analog integrated circuits, digitally-assisted analog circuits, and VLSI computer-aided design.

Dr. Iizuka is a member of the Institute of Electronics, Information and Communication Engineers (IEICE). He was a member of the IEEE International Solid-State Circuits Conference (ISSCC) Technical Program Committee from 2013 to 2017 and a member of the IEEE Custom Integrated Circuits Conference (CICC) Technical Program Committee from 2014 to 2019. From 2016 to 2018, he served as the Editor for *IEICE Electronics Express* (ELEX). He is currently serving as a member for the IEEE Asian Solid-State Circuits Conference (A-SSCC) and IEEE VLSI Symposium on Circuits Technical Program Committees. He was a recipient of the 21st Marubun Research Encouragement Commendation from Marubun Research Promotion Foundation in 2018, the 13th Wakashachi Encouragement Award First Prize in 2019, and the 18th Funai Academic Prize from Funai Foundation for Information Technology in 2019. He was also a co-recipient of the IEEE International Test Conference Ned Kornfield Best Paper Award in 2016.